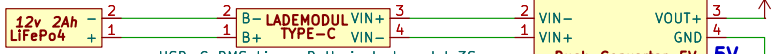
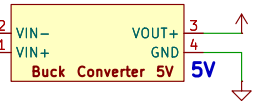
AKKU + POWER SOURCE

U7
12V 2Ah LiFePo4 Akku mit BMS



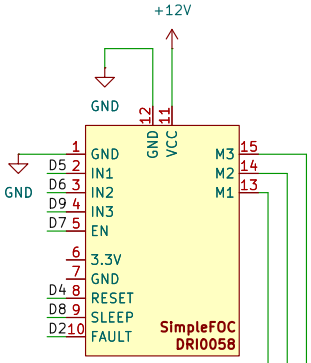
USB-C BMS Li-on Batterie Lademodul 3S
U8

Waveshare DC-DC Buck Mini Module
U6



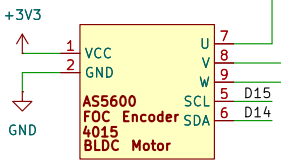
+5V
5V
GND

MOTOR + TREIBER



U3
SIMPLEFOCMINI BRUSHLESS DC MOTOR - DRI0058

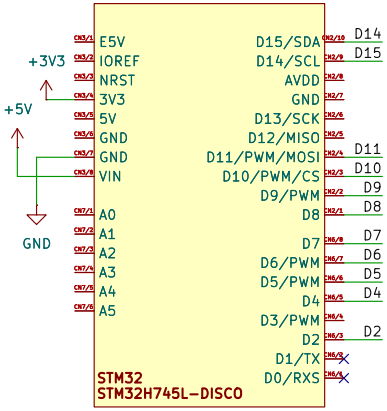
U2



4015 BLDC Motor + AS5600 FOC Encoder (Integriert am Motor)

uC

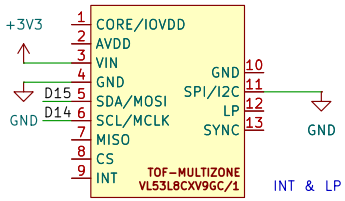
U1
STM32H745I-Discovery Board (Arduino Header)



D1 & D0 reserviert für USART

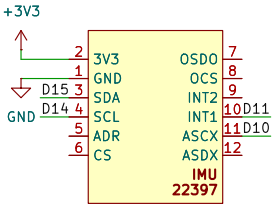
TOF SENSOR

U5 TOF-MULTIZONE - VL53L8CXV9GC/1



INT & LP später umsetzen

IMU SENSOR



SPARKFUN 6DOF IMU BREAKOUT - 22397
U4

Jethro Noah Wangge		
Sheet: /		
File: kiCAD_Bachelorarbeit.kicad_sch		
Title: Bachelorarbeit_Virtueller_Sicherheitsbereich_am_Rad		
Size: A4	Date: 2026-08-14	Rev: B
KiCad E.D.A. 10.0.3		Id: 1/1